

Class 09: Candy Mini-Project

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Background

Today we are taking a detour to analyze a fun dataset (that we have more intrinsic insight into) with the most useful analysis method we have learned thus far - Principal Component Analysis (PCA)

Data Import

The data is all related to halloween candy and is from the 538 website.

```
candy <- read.csv ("candy-data.txt", row.names = 1)
head(candy)
```

	chocolate	fruity	caramel	peanut	almond	nougat	crisped	rice	wafer
100 Grand	1	0	1		0	0			1
3 Musketeers	1	0	0		0	1			0
One dime	0	0	0		0	0			0
One quarter	0	0	0		0	0			0

Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0
	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

Q1. How many different candy types are in this dataset?

```
nrow(candy)
```

```
[1] 85
```

Q2. How many fruity candy types are in the dataset?

```
sum(candy$fruity)
```

```
[1] 38
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is its winpercent value?

```
library(dplyr)
```

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':
```

```
filter, lag
```

```
The following objects are masked from 'package:base':
```

```
intersect, setdiff, setequal, union
```

```
candy |>
  filter(row.names(candy)=="Almond Joy") |>
  select(winpercent)
```

```
      winpercent
Almond Joy  50.34755
```

```
# filter() selects rows that match a condition, specifically selecting a
#specific candy shown here
```

```
# select() chooses specific columns from the dataset
```

Q4. What is the winpercent value for “Kit Kat”?

```
candy |>
  filter(row.names(candy)=="Kit Kat") |>
  select(winpercent)
```

```
      winpercent
Kit Kat    76.7686
```

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

```
candy |>
  filter(row.names(candy)=="Tootsie Roll Snack Bars") |>
  select(winpercent)
```

```
      winpercent
Tootsie Roll Snack Bars  49.6535
```

There is a useful `skim()` function in the **skimr** package that can help give you a quick overview of a given dataset. Let’s install this package and try it on our candy data.

```
library("skimr")
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency: numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

```
# The skim() function provides a summary of the dataset by grouping variables
# based on their data type.
```

```
# skim(candy) scans through the dataset and organizes the output by variable
# type.
```

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

Yes, the winpercent variable is on a different scale in comparison to the majority of other columns. Even though most variables are binary 0 or 1, winpercent is a continuous variable representing a percentage. Additionally, the values range from 0 to 100.

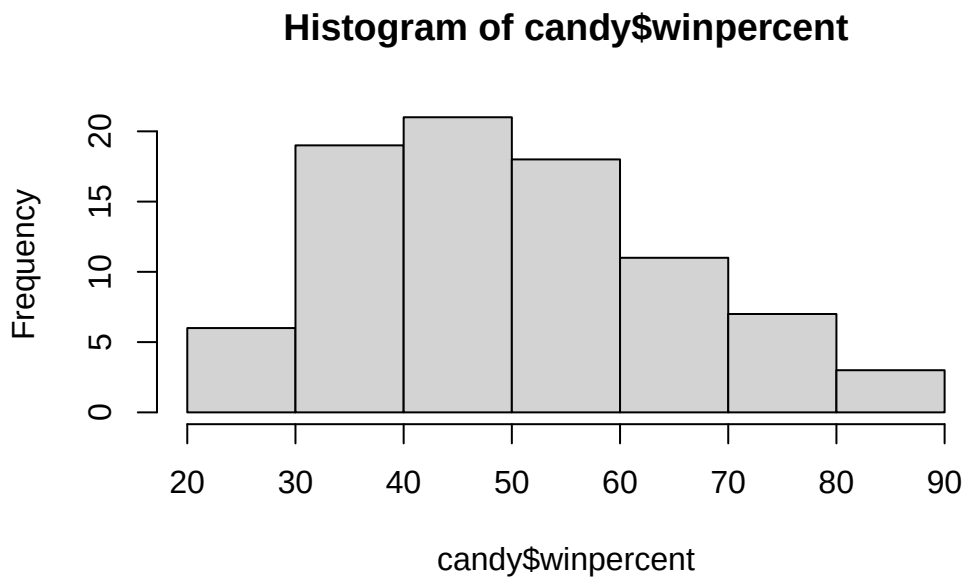
Q7. What do you think a zero and one represent for the `candy$chocolate` column?

In the `candy$chocolate` column, a value of 1 indicates that the candy contains chocolate, while a value of 0 indicates that it does not.

Exploratory Analysis

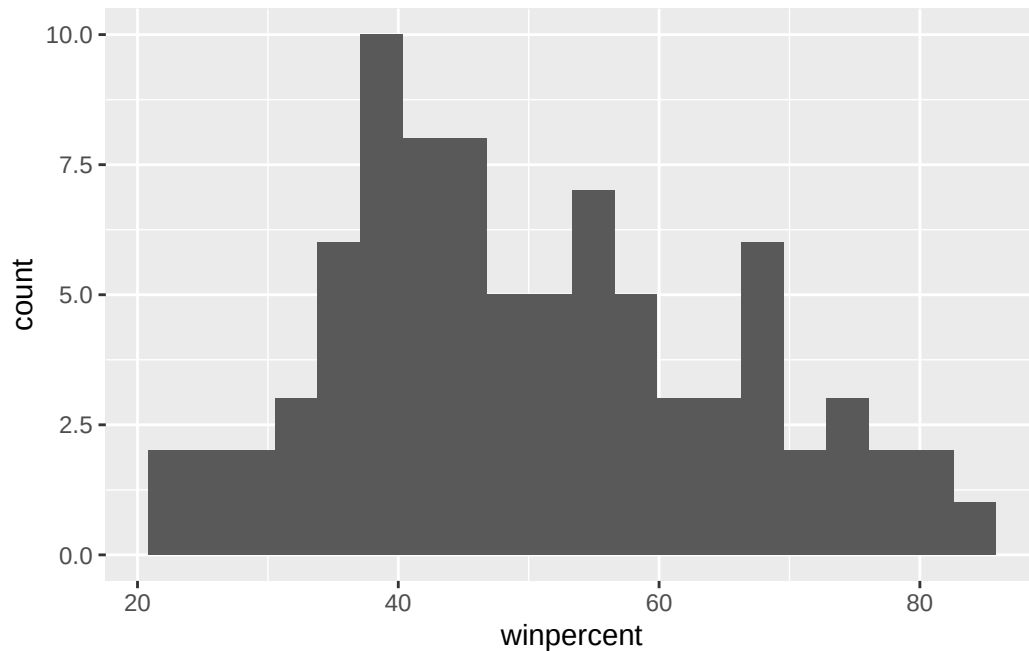
Q8. Plot a histogram of `winpercent` values using both base R and `ggplot2`.

```
hist(candy$winpercent)
```



```
library(ggplot2)

ggplot(candy) +
  aes(winpercent) +
  geom_histogram(bins=20)
```



Q9. Is the distribution of winpercent values symmetrical?

No, the distribution of the winpercent values skew to the right.

Q10. Is the center of the distribution above or below 50%?

```
mean(candy$winpercent)
```

```
[1] 50.31676
```

```
summary(candy$winpercent)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
22.45	39.14	47.83	50.32	59.86	84.18

The center of the distribution is below 50% since the median winpercent is 47.83%. Even though the mean is slightly above 50%, the median better represents the center because the distribution is slightly skewed.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
choc.ind <- as.logical(candy$chocolate)
choc.candy <- candy[choc.ind,]
choc.win <- choc.candy$winpercent
choc.win
```

```
[1] 66.97173 67.60294 50.34755 56.91455 38.97504 55.37545 62.28448 56.49050
[9] 59.23612 57.21925 76.76860 71.46505 66.57458 55.06407 73.09956 60.80070
[17] 64.35334 47.82975 54.52645 70.73564 66.47068 69.48379 81.86626 84.18029
[25] 73.43499 72.88790 65.71629 34.72200 37.88719 76.67378 59.52925 48.98265
[33] 43.06890 45.73675 49.65350 81.64291 49.52411
```

```
mean(choc.win)
```

```
[1] 60.92153
```

```
fruit.ind <- as.logical(candy$fruity)
# converts the chocolate column into True/False values
fruit.candy <- candy[fruit.ind,]
# subsets the dataset to only include chocolate candies
fruit.win <- fruit.candy$winpercent
# extracts the winpercent values for chocolate candies
mean(fruit.win)
```

```
[1] 44.11974
```

```
# this calculates the average winpercent for chocolate candies
```

```
candy |>
  group_by(chocolate) |>
  # group_by() splits the data into groups (such as chocolate versus non
  # chocolate)
  summarise(avg_win = mean(winpercent))
```

```
# A tibble: 2 x 2
  chocolate avg_win
    <int>    <dbl>
1       0     42.1
2       1     60.9
```

```
# summarise() calculates a summary statistic (mean winpercent) for each group
```

On average, chocolate candy is ranked higher than fruit candy, as candies containing chocolate have a higher mean winpercent.

Q12. Is this difference statistically significant?

```
t.test(choc.win, fruit.win)
```

Welch Two Sample t-test

```
data:  choc.win and fruit.win
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

Yes, the difference is statistically significant since the p-value, 2.871e-08, from the t-test is less than 0.05. This suggests that the difference in average winpercent between chocolate candies and fruity candies is unlikely to be due to chance.

Overall Candy Rankings

Q13. What are the five least liked candy types in this set?

```
candy |>
  arrange(winpercent) |>
  # arrange() sorts the data in ascending order by winpercent
  select(winpercent) |>
  head(5)
```

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

The 5 least liked candy are Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble and Jawbusters.

```
x <- c(5,10,1)
sort(x)
```

```
[1] 1 5 10
```

```
order(x)
```

```
[1] 3 1 2
```

```
candy[ order(candy$winpercent), ]
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Nik L Nip	0	1	0	0	0
Boston Baked Beans	0	0	0	1	0
Chiclets	0	1	0	0	0
Super Bubble	0	1	0	0	0
Jawbusters	0	1	0	0	0
Root Beer Barrels	0	0	0	0	0
Sugar Daddy	0	0	1	0	0
One dime	0	0	0	0	0
Sugar Babies	0	0	1	0	0
Haribo Happy Cola	0	0	0	0	0
Caramel Apple Pops	0	1	1	0	0
Strawberry bon bons	0	1	0	0	0
Sixlets	1	0	0	0	0
Ring pop	0	1	0	0	0
Chewey Lemonhead Fruit Mix	0	1	0	0	0
Red vines	0	1	0	0	0
Pixie Sticks	0	0	0	0	0
Nestle Smarties	1	0	0	0	0
Candy Corn	0	0	0	0	0
Charleston Chew	1	0	0	0	1
Warheads	0	1	0	0	0
Lemonhead	0	1	0	0	0
Fun Dip	0	1	0	0	0
Now & Later	0	1	0	0	0
Dum Dums	0	1	0	0	0

Pop Rocks	0	1	0	0	0
Laffy Taffy	0	1	0	0	0
Werther's Original Caramel	0	0	1	0	0
Haribo Twin Snakes	0	1	0	0	0
Dots	0	1	0	0	0
Runts	0	1	0	0	0
Tootsie Roll Juniors	1	0	0	0	0
Fruit Chews	0	1	0	0	0
Welch's Fruit Snacks	0	1	0	0	0
Twizzlers	0	1	0	0	0
Tootsie Roll Midgies	1	0	0	0	0
Smarties candy	0	1	0	0	0
One quarter	0	0	0	0	0
Payday	0	0	0	1	1
Mike & Ike	0	1	0	0	0
Gobstopper	0	1	0	0	0
Trolli Sour Bites	0	1	0	0	0
Mounds	1	0	0	0	0
Tootsie Pop	1	1	0	0	0
Whoppers	1	0	0	0	0
Tootsie Roll Snack Bars	1	0	0	0	0
Almond Joy	1	0	0	1	0
Haribo Sour Bears	0	1	0	0	0
Air Heads	0	1	0	0	0
Sour Patch Tricksters	0	1	0	0	0
Lifesavers big ring gummies	0	1	0	0	0
Mr Good Bar	1	0	0	1	0
Swedish Fish	0	1	0	0	0
Milk Duds	1	0	1	0	0
Skittles wildberry	0	1	0	0	0
Nerds	0	1	0	0	0
Hershey's Kisses	1	0	0	0	0
Hershey's Milk Chocolate	1	0	0	0	0
Baby Ruth	1	0	1	1	1
Haribo Gold Bears	0	1	0	0	0
Junior Mints	1	0	0	0	0
Hershey's Special Dark	1	0	0	0	0
Snickers Crisper	1	0	1	1	0
Sour Patch Kids	0	1	0	0	0
Milky Way Midnight	1	0	1	0	1
Hershey's Krackel	1	0	0	0	0
Skittles original	0	1	0	0	0
Milky Way Simply Caramel	1	0	1	0	0

Rolo	1	0	1	0	0
Nestle Crunch	1	0	0	0	0
M&M's	1	0	0	0	0
100 Grand	1	0	1	0	0
Starburst	0	1	0	0	0
3 Musketeers	1	0	0	0	1
Peanut M&Ms	1	0	0	1	0
Nestle Butterfinger	1	0	0	1	0
Peanut butter M&M's	1	0	0	1	0
Reese's stuffed with pieces	1	0	0	1	0
Milky Way	1	0	1	0	1
Reese's pieces	1	0	0	1	0
Snickers	1	0	1	1	1
Kit Kat	1	0	0	0	0
Twix	1	0	1	0	0
Reese's Miniatures	1	0	0	1	0
Reese's Peanut Butter cup	1	0	0	1	0
	crisped	rice	wafer	hard	bar
	pluribus	sugar	percent		
Nik L Nip	0	0	0	1	0.197
Boston Baked Beans	0	0	0	1	0.313
Chiclets	0	0	0	1	0.046
Super Bubble	0	0	0	0	0.162
Jawbusters	0	1	0	1	0.093
Root Beer Barrels	0	1	0	1	0.732
Sugar Daddy	0	0	0	0	0.418
One dime	0	0	0	0	0.011
Sugar Babies	0	0	0	1	0.965
Haribo Happy Cola	0	0	0	1	0.465
Caramel Apple Pops	0	0	0	0	0.604
Strawberry bon bons	0	1	0	1	0.569
Sixlets	0	0	0	1	0.220
Ring pop	0	1	0	0	0.732
Chewey Lemonhead Fruit Mix	0	0	0	1	0.732
Red vines	0	0	0	1	0.581
Pixie Sticks	0	0	0	1	0.093
Nestle Smarties	0	0	0	1	0.267
Candy Corn	0	0	0	1	0.906
Charleston Chew	0	0	1	0	0.604
Warheads	0	1	0	0	0.093
Lemonhead	0	1	0	0	0.046
Fun Dip	0	1	0	0	0.732
Now & Later	0	0	0	1	0.220
Dum Dums	0	1	0	0	0.732

Pop Rocks	0	1	0	1	0.604
Laffy Taffy	0	0	0	0	0.220
Werther's Original Caramel	0	1	0	0	0.186
Haribo Twin Snakes	0	0	0	1	0.465
Dots	0	0	0	1	0.732
Runts	0	1	0	1	0.872
Tootsie Roll Juniors	0	0	0	0	0.313
Fruit Chews	0	0	0	1	0.127
Welch's Fruit Snacks	0	0	0	1	0.313
Twizzlers	0	0	0	0	0.220
Tootsie Roll Midgies	0	0	0	1	0.174
Smarties candy	0	1	0	1	0.267
One quarter	0	0	0	0	0.011
Payday	0	0	1	0	0.465
Mike & Ike	0	0	0	1	0.872
Gobstopper	0	1	0	1	0.906
Trolli Sour Bites	0	0	0	1	0.313
Mounds	0	0	1	0	0.313
Tootsie Pop	0	1	0	0	0.604
Whoppers	1	0	0	1	0.872
Tootsie Roll Snack Bars	0	0	1	0	0.465
Almond Joy	0	0	1	0	0.465
Haribo Sour Bears	0	0	0	1	0.465
Air Heads	0	0	0	0	0.906
Sour Patch Tricksters	0	0	0	1	0.069
Lifesavers big ring gummies	0	0	0	0	0.267
Mr Good Bar	0	0	1	0	0.313
Swedish Fish	0	0	0	1	0.604
Milk Duds	0	0	0	1	0.302
Skittles wildberry	0	0	0	1	0.941
Nerds	0	1	0	1	0.848
Hershey's Kisses	0	0	0	1	0.127
Hershey's Milk Chocolate	0	0	1	0	0.430
Baby Ruth	0	0	1	0	0.604
Haribo Gold Bears	0	0	0	1	0.465
Junior Mints	0	0	0	1	0.197
Hershey's Special Dark	0	0	1	0	0.430
Snickers Crisper	1	0	1	0	0.604
Sour Patch Kids	0	0	0	1	0.069
Milky Way Midnight	0	0	1	0	0.732
Hershey's Krackel	1	0	1	0	0.430
Skittles original	0	0	0	1	0.941
Milky Way Simply Caramel	0	0	1	0	0.965

Rolo	0	0	0	1	0.860
Nestle Crunch	1	0	1	0	0.313
M&M's	0	0	0	1	0.825
100 Grand	1	0	1	0	0.732
Starburst	0	0	0	1	0.151
3 Musketeers	0	0	1	0	0.604
Peanut M&Ms	0	0	0	1	0.593
Nestle Butterfinger	0	0	1	0	0.604
Peanut butter M&M's	0	0	0	1	0.825
Reese's stuffed with pieces	0	0	0	0	0.988
Milky Way	0	0	1	0	0.604
Reese's pieces	0	0	0	1	0.406
Snickers	0	0	1	0	0.546
Kit Kat	1	0	1	0	0.313
Twix	1	0	1	0	0.546
Reese's Miniatures	0	0	0	0	0.034
Reese's Peanut Butter cup	0	0	0	0	0.720

	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Boston Baked Beans	0.511	23.41782
Chiclets	0.325	24.52499
Super Bubble	0.116	27.30386
Jawbusters	0.511	28.12744
Root Beer Barrels	0.069	29.70369
Sugar Daddy	0.325	32.23100
One dime	0.116	32.26109
Sugar Babies	0.767	33.43755
Haribo Happy Cola	0.465	34.15896
Caramel Apple Pops	0.325	34.51768
Strawberry bon bons	0.058	34.57899
Sixlets	0.081	34.72200
Ring pop	0.965	35.29076
Chewey Lemonhead Fruit Mix	0.511	36.01763
Red vines	0.116	37.34852
Pixie Sticks	0.023	37.72234
Nestle Smarties	0.976	37.88719
Candy Corn	0.325	38.01096
Charleston Chew	0.511	38.97504
Warheads	0.116	39.01190
Lemonhead	0.104	39.14106
Fun Dip	0.325	39.18550
Now & Later	0.325	39.44680
Dum Dums	0.034	39.46056

Pop Rocks	0.837	41.26551
Laffy Taffy	0.116	41.38956
Werther's Original Caramel	0.267	41.90431
Haribo Twin Snakes	0.465	42.17877
Dots	0.511	42.27208
Runts	0.279	42.84914
Tootsie Roll Juniors	0.511	43.06890
Fruit Chews	0.034	43.08892
Welch's Fruit Snacks	0.313	44.37552
Twizzlers	0.116	45.46628
Tootsie Roll Midgies	0.011	45.73675
Smarties candy	0.116	45.99583
One quarter	0.511	46.11650
Payday	0.767	46.29660
Mike & Ike	0.325	46.41172
Gobstopper	0.453	46.78335
Trolli Sour Bites	0.255	47.17323
Mounds	0.860	47.82975
Tootsie Pop	0.325	48.98265
Whoppers	0.848	49.52411
Tootsie Roll Snack Bars	0.325	49.65350
Almond Joy	0.767	50.34755
Haribo Sour Bears	0.465	51.41243
Air Heads	0.511	52.34146
Sour Patch Tricksters	0.116	52.82595
Lifesavers big ring gummies	0.279	52.91139
Mr Good Bar	0.918	54.52645
Swedish Fish	0.755	54.86111
Milk Duds	0.511	55.06407
Skittles wildberry	0.220	55.10370
Nerds	0.325	55.35405
Hershey's Kisses	0.093	55.37545
Hershey's Milk Chocolate	0.918	56.49050
Baby Ruth	0.767	56.91455
Haribo Gold Bears	0.465	57.11974
Junior Mints	0.511	57.21925
Hershey's Special Dark	0.918	59.23612
Snickers Crisper	0.651	59.52925
Sour Patch Kids	0.116	59.86400
Milky Way Midnight	0.441	60.80070
Hershey's Krackel	0.918	62.28448
Skittles original	0.220	63.08514
Milky Way Simply Caramel	0.860	64.35334

Rolo	0.860	65.71629
Nestle Crunch	0.767	66.47068
M&M's	0.651	66.57458
100 Grand	0.860	66.97173
Starburst	0.220	67.03763
3 Musketeers	0.511	67.60294
Peanut M&Ms	0.651	69.48379
Nestle Butterfinger	0.767	70.73564
Peanut butter M&M's	0.651	71.46505
Reese's stuffed with pieces	0.651	72.88790
Milky Way	0.651	73.09956
Reese's pieces	0.651	73.43499
Snickers	0.651	76.67378
Kit Kat	0.511	76.76860
Twix	0.906	81.64291
Reese's Miniatures	0.279	81.86626
Reese's Peanut Butter cup	0.651	84.18029

Q14. What are the top 5 all time favorite candy types out of this set?

```
candy |>
  arrange(desc(winpercent)) |>
  # desc() sorts in descending order, highest first
  head(5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Reese's Peanut Butter cup	1	0	0		1	0
Reese's Miniatures	1	0	0		1	0
Twix	1	0	1		0	0
Kit Kat	1	0	0		0	0
Snickers	1	0	1		1	1

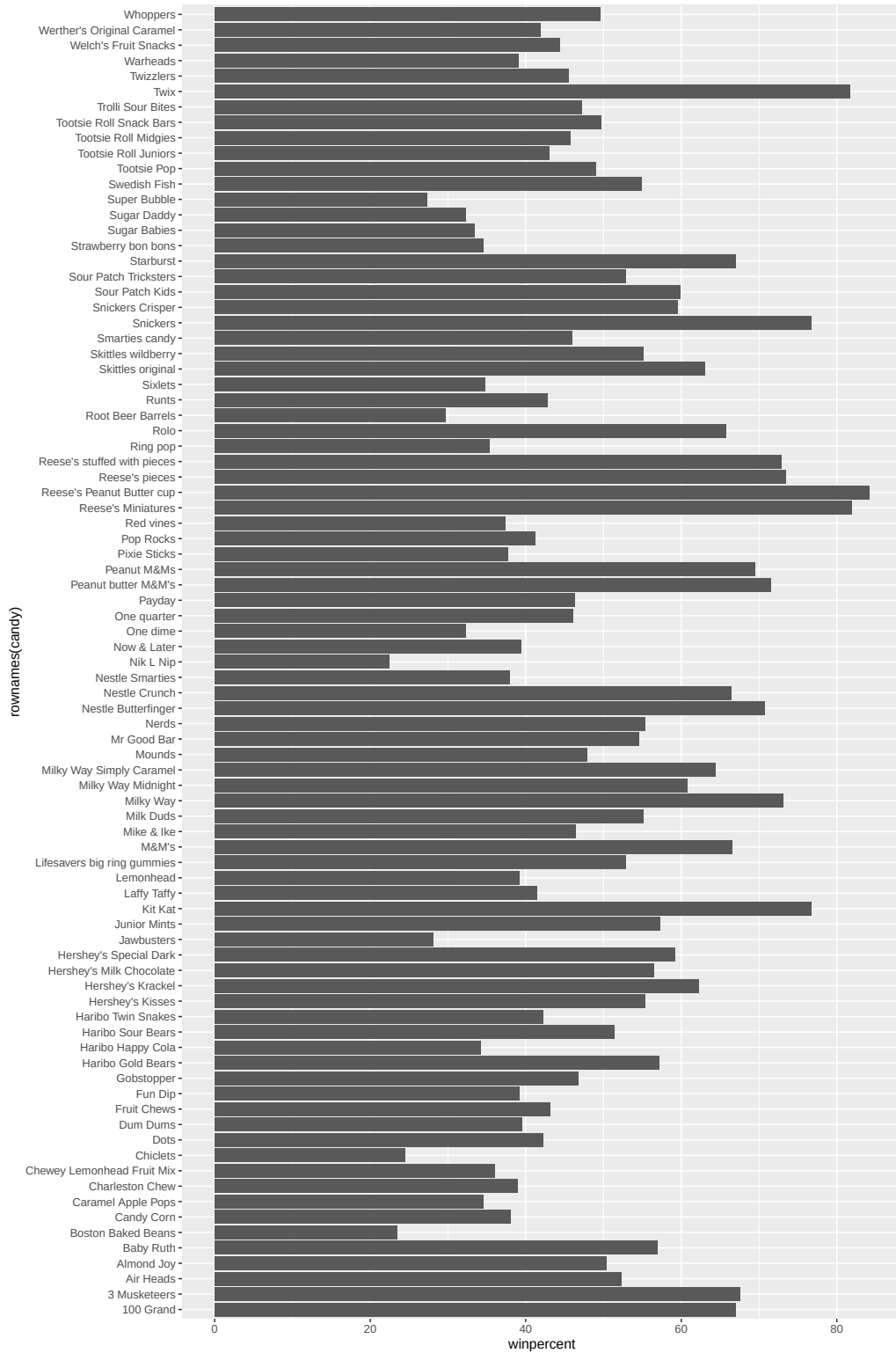
	crisp	rice	waffer	hard bar	pluribus	sugar
Reese's Peanut Butter cup		0	0	0	0	0.720
Reese's Miniatures		0	0	0	0	0.034
Twix		1	0	1	0	0.546
Kit Kat		1	0	1	0	0.313
Snickers		0	0	1	0	0.546

	price	percent	winpercent
Reese's Peanut Butter cup	0.651		84.18029
Reese's Miniatures	0.279		81.86626
Twix	0.906		81.64291
Kit Kat	0.511		76.76860
Snickers	0.651		76.67378

The top 5 favorite candies are Reese's Peanut Butter cup, Reese's Miniatures, Twix, Kit Kat and Snickers.

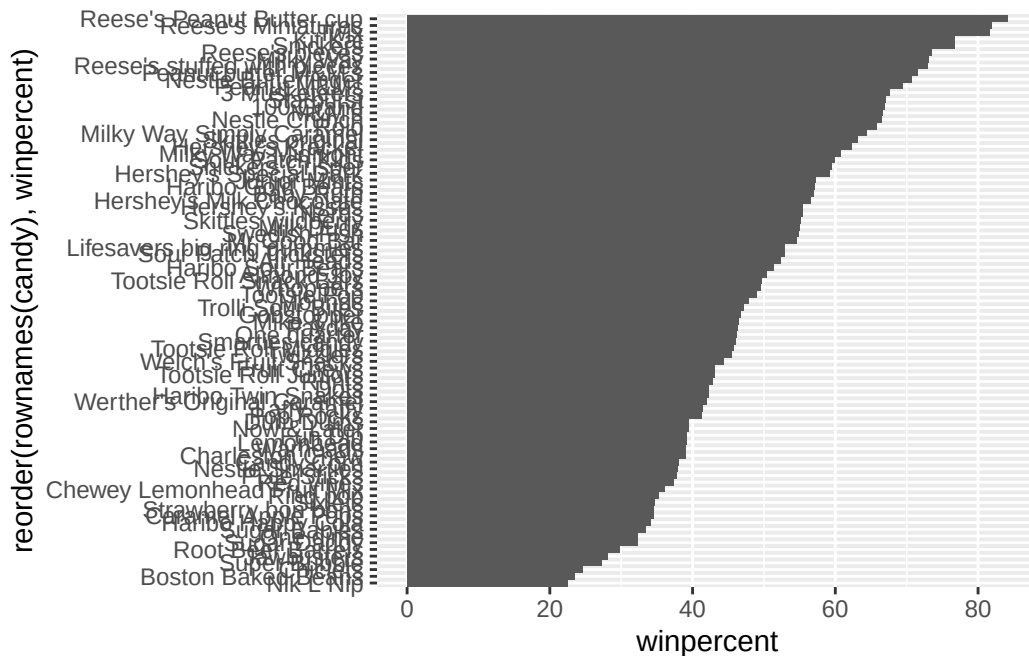
Q15. Make a firstbarplot of candy ranking based o winpercent values.

```
ggplot(candy) +  
  aes(winpercent, rownames(candy)) +  
  geom_col()
```

Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by `winpercent`?

```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col()
```



```
# reorder() rearranges factor levels based on another variable, sorts bars by
# winpercent
# geom_col() creates a bar plot using actual values
```

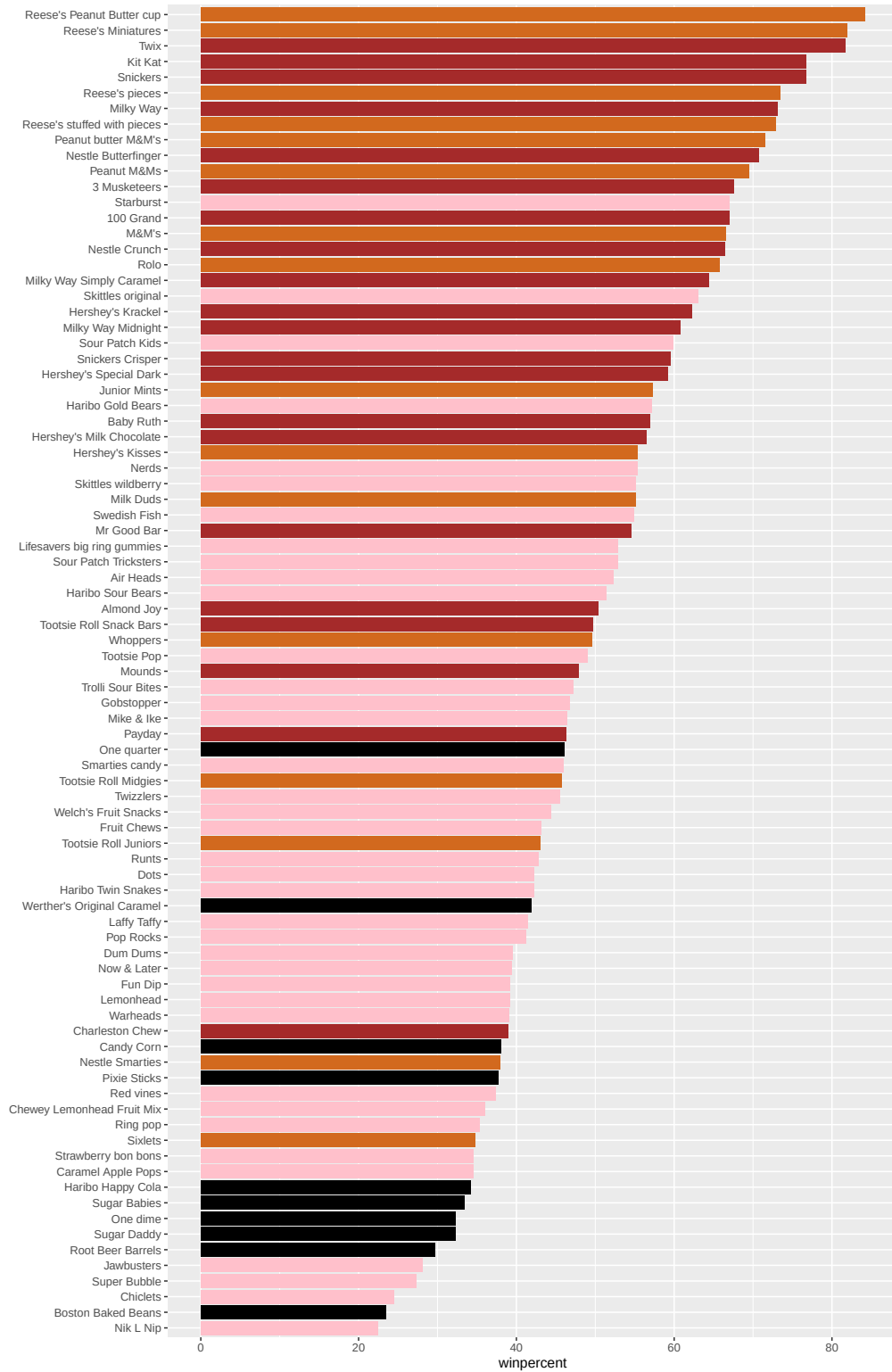
```
mycols <- rep("black", nrow(candy))
# creates a color vector with one color for each candy

# Chocolate candy
mycols[as.logical(candy$chocolate)] <- "chocolate"
# colors the chocolate candies

# Candy bars in brown
mycols [as.logical(candy$bar)] <- "brown"
# the candy bar is colored brown
```

```
# Fruity candy in pink
mycols [as.logical(candy$fruity)] <- "pink"
# this colors fruity candies in pink

ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=mycols) +
  ylab("")
```



Q17. What is the worst ranked chocolate candy?

```
candy |>
  filter(chocolate == 1) |>
  arrange(winpercent) |>
  select(winpercent) |>
  head(1)
```

	winpercent
Sixlets	34.722

Sixlets is the worst ranked chocolate candy.

Q18. What is the best ranked fruity candy?

```
candy |>
  filter(fruity == 1) |>
  arrange(desc(winpercent)) |>
  select(winpercent) |>
  head(1)
```

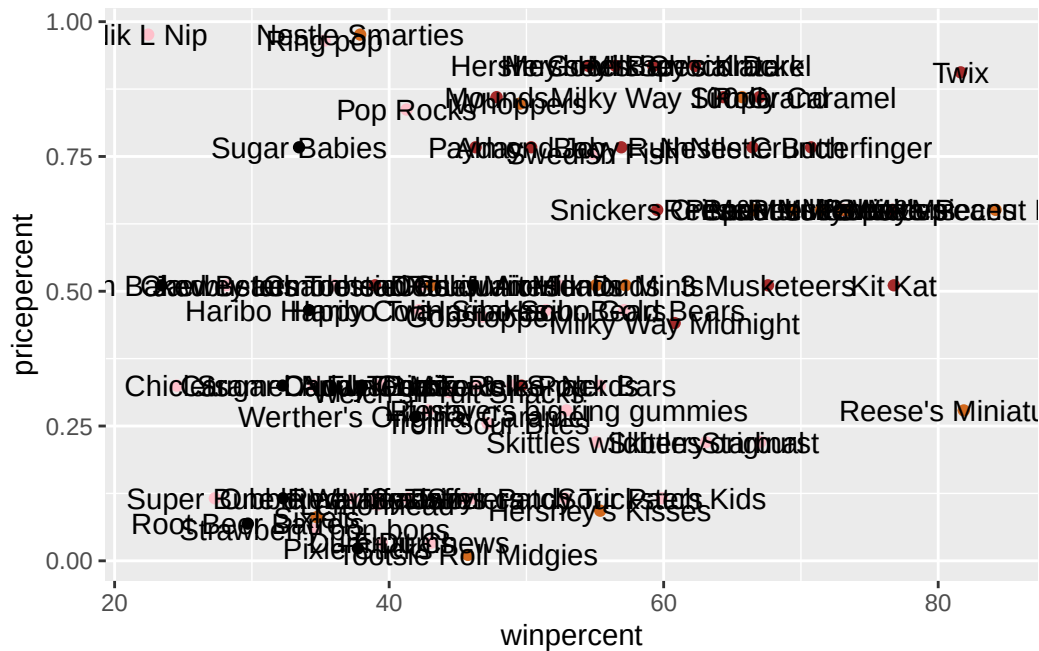
	winpercent
Starburst	67.03763

Starburst is the best ranked fruity candy.

Taking a look at pricepercent

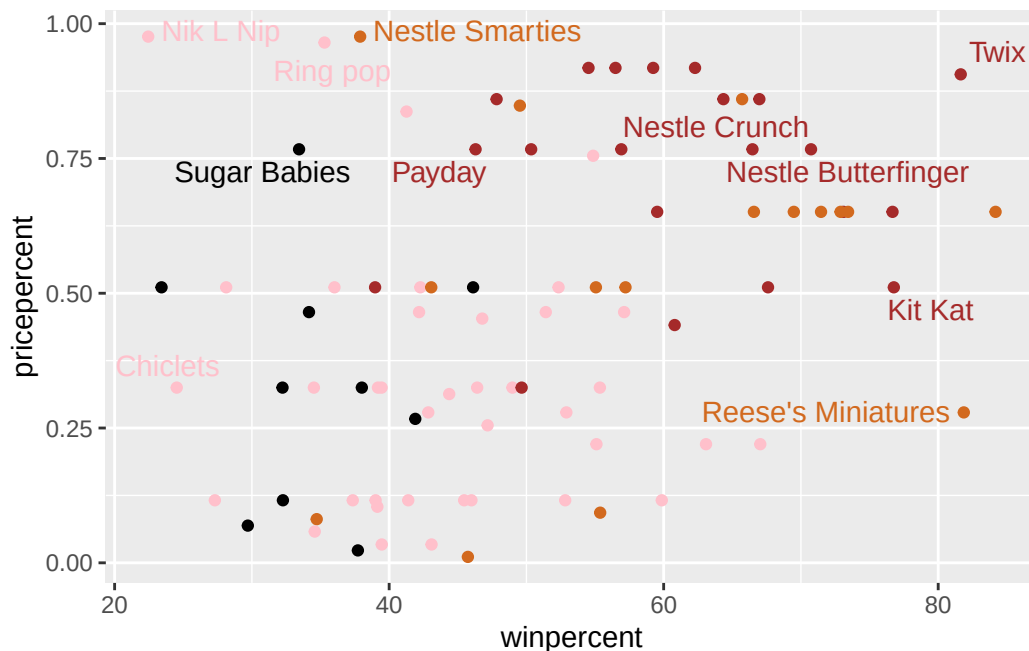
Make a plot of winpercent vs pricepercent

```
ggplot(candy) +
  aes(winpercent,
      pricepercent,
      label=rownames(candy)) +
  geom_point(col=mycols) +
  geom_text()
```



We can fix the label text overplotting with an add-on package called **ggrepel** and it's `geom_text_repel()`

```
library (ggrepel)
ggplot(candy) +
  aes(winpercent,
       pricepercent,
       label=rownames(candy)) +
  geom_point(col=mycols) +
  geom_text_repel(col=mycols, max.overlaps = 5)
```



```
# geom_text_repel() adds labels to points while preventing overlapping text
```

Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

The Reese's Miniature is the highest ranked in terms of winpercent for the least money.

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
ord <- order(candy$pricepercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )
```

	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

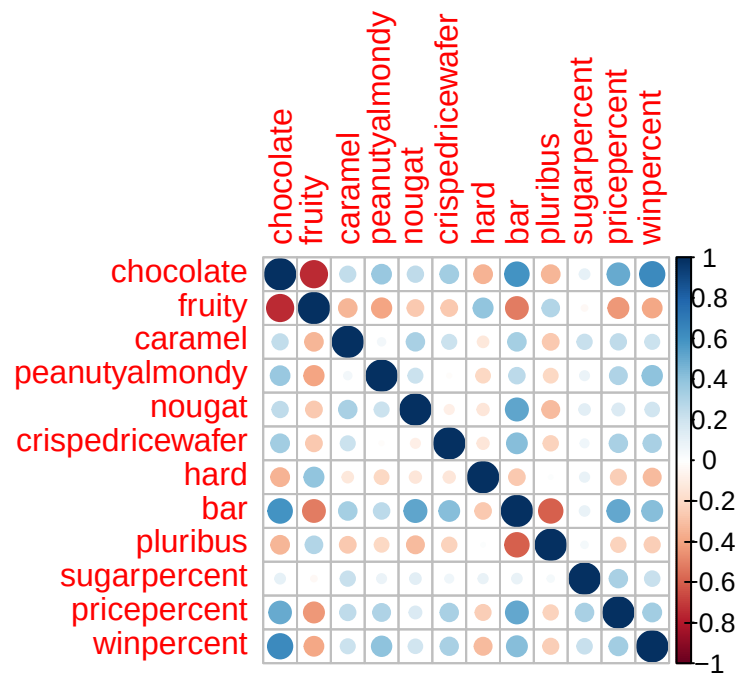
The top 5 most expensive candy types in the dataset are Nik L Nip, Nestle Smarties, Ring pop, Hershey's Krackel, and Hershey's Milk Chocolate. The least popular is Nik L Nip since it has the lowest winpercent among them.

Exploring the Correlation Structure

```
library(corrplot)
```

```
corrplot 0.95 loaded
```

```
cij <- cor(candy)  
corrplot(cij)
```



```
# visualizes the correlation matrix with colors representing the strength and  
# direction
```

Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

The two variables “Chocolate” and “Fruity” are anti-correlated because they have red values for each other.

Q23. Use your corrplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

I expect chocolate and bar to have large positive contributions to PC1, while fruity will have a large negative contribution since chocolate and bar are strongly positively correlated, while fruity is strongly anti-correlated with chocolate. Additionally, variables like pricepercent and winpercent could also contribute to PC1 since they show positive relationships with chocolate. Hence, this suggests that PC1 will mainly separate chocolate-based candies from fruity candies as a result.

Principal Component Analysis

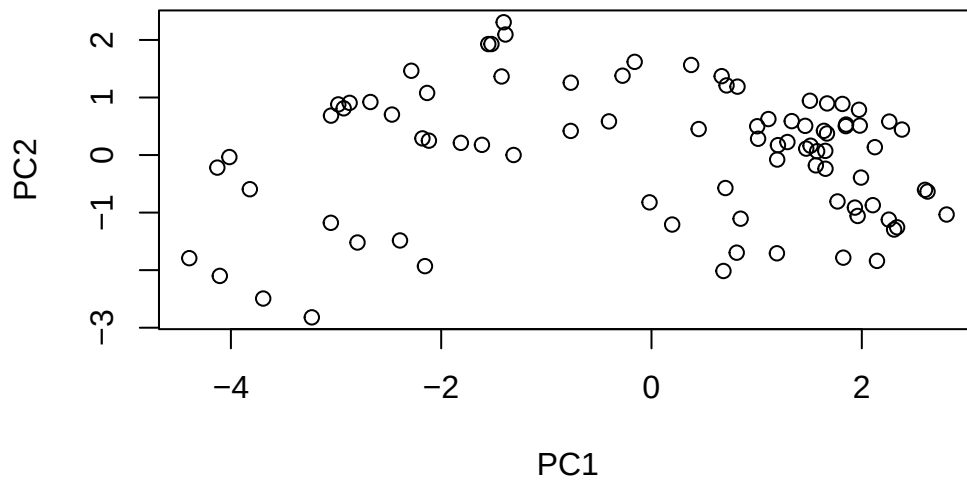
```
pca <- prcomp(candy, scale=TRUE)
summary(pca)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

```
plot(pca$x[,1:2])
```



```
# prcomp() performs principal component analysis
# scale=TRUE standardizes the variables so they contribute equally
```

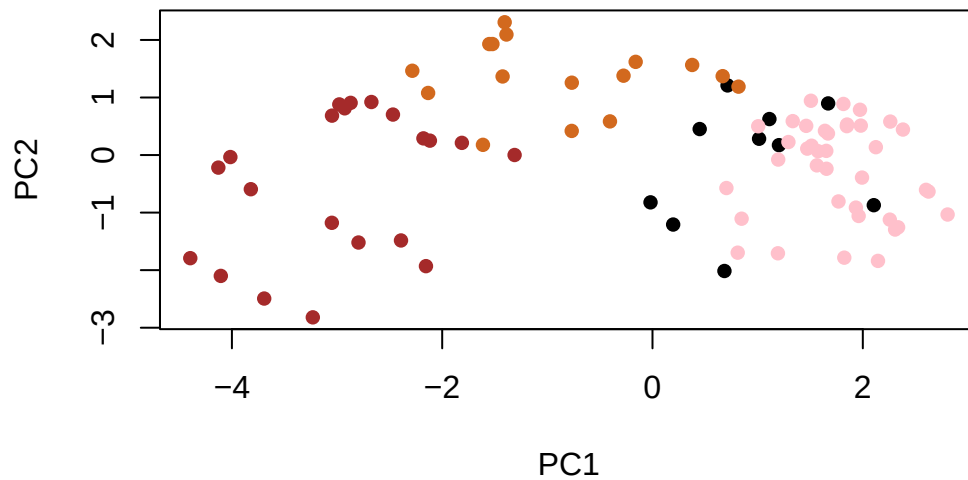
```
pca <- prcomp(candy, scale=TRUE)
summary(pca)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

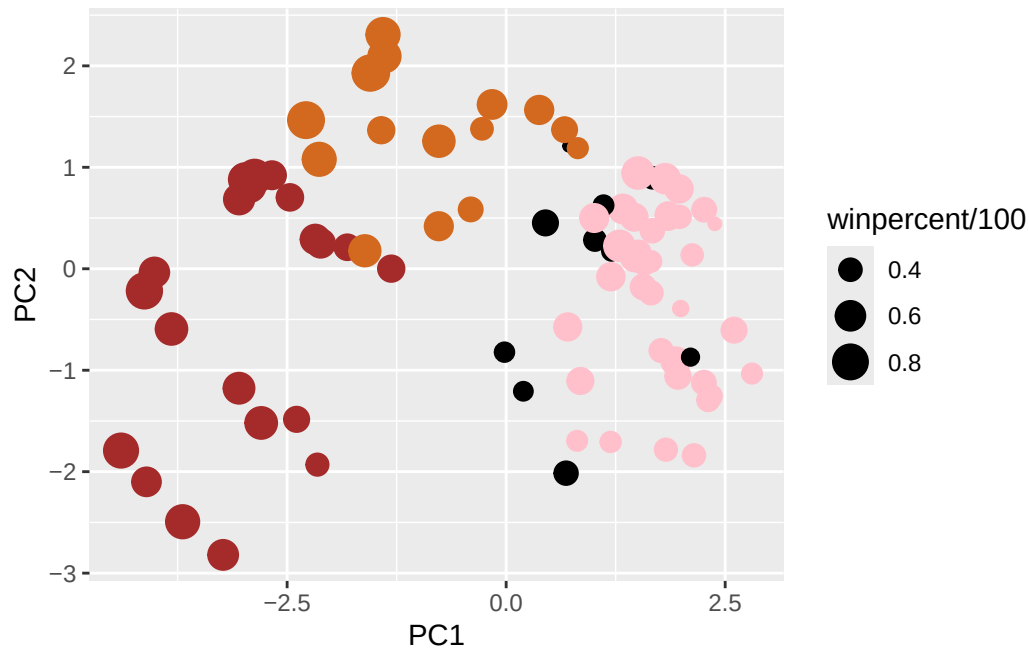
	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

```
plot(pca$x[,1:2], col=mycols, pch=16)
```



```
# Make a new data-frame with our PCA results and candy data
my_data <- cbind(candy, pca$x[,1:3])
#cbind() combines the datasets columnwise so it adds PCA results to original
# data
```

```
p <- ggplot(my_data) +
  aes(x=PC1, y=PC2,
      size=winpercent/100,
      text=rownames(my_data),
      label=rownames(my_data)) +
  geom_point(col=mycols)
p
```



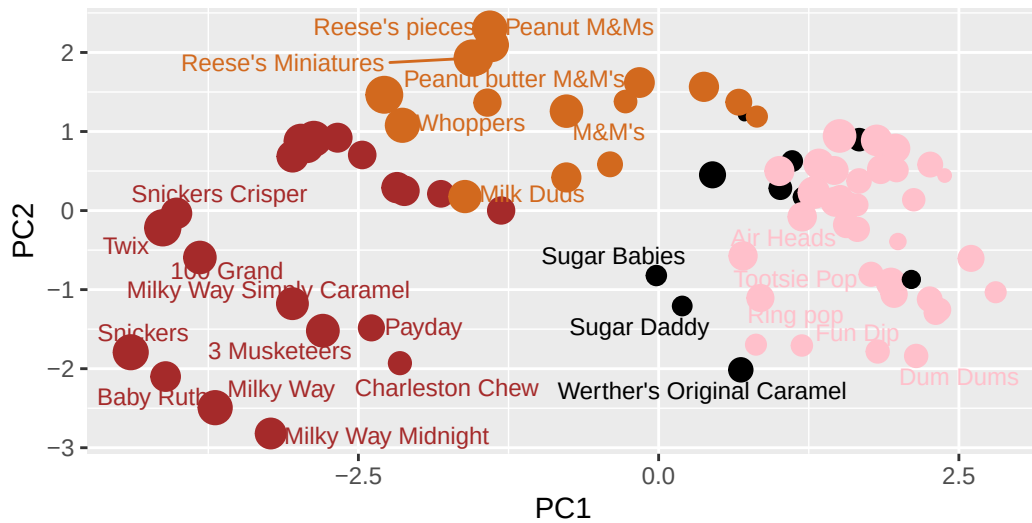
```
library(ggrepel)
# loads the ggrepel package and improves text labeling in ggplot

p + geom_text_repel(size=3.3, col=mycols, max.overlaps = 7) +
# geom_text_repel() adds labels to each point while minimizing overlap between
# labels

# max.overlaps limits how many labels can overlap
theme(legend.position = "none") +
labs(title="Halloween Candy PCA Space",
      subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),
      caption="Data from 538")
```

Halloween Candy PCA Space

Colored by type: chocolate bar (dark brown), chocolate other (light brown),



Data from 538

```
# labs() adds descriptive labels to the plot
```

I expect fruity, chocolate and bar candies to have the largest contributions to PC1 since chocolate and bar should load together seeing as they are positively correlated. In comparison, fruity should load in the opposite direction since fruity and chocolate are anti-correlated. Overall, this suggests that PC1 will separate chocolate bar candies from fruity candies. ## Principal Component Analysis

In this case we want to be sure to set `scale=TRUE` argument for `prcomp()` because we have one variable `winpercent` that is on a very different scale than all others and would otherwise dominate our PCA results.

```
pca <- prcomp(candy, scale=TRUE)
summary(pca)
```

Importance of components:

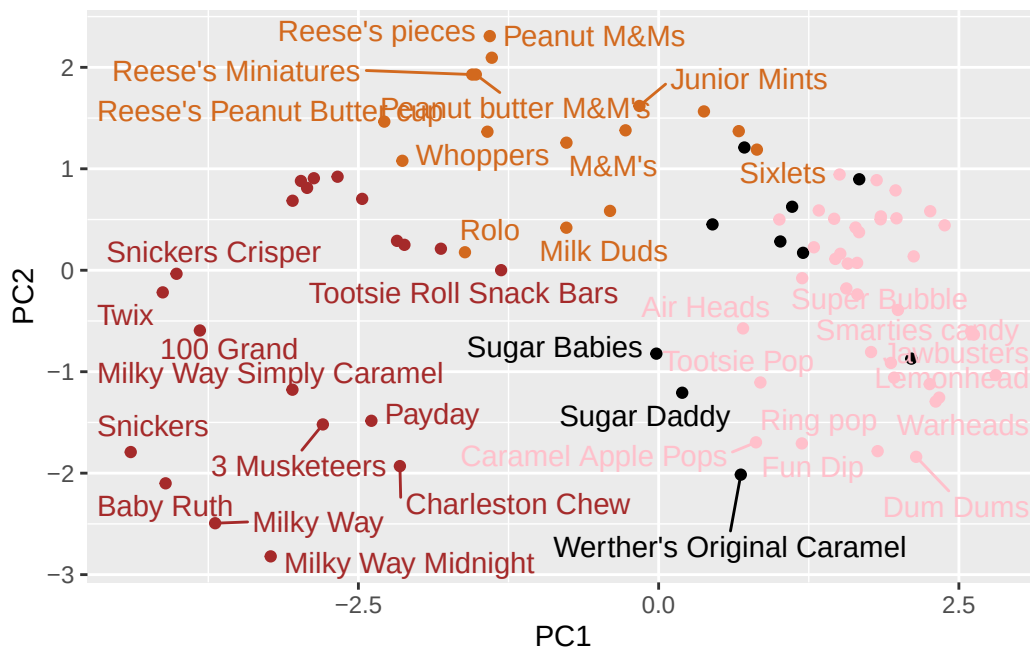
	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760

Proportion of Variance 0.04629 0.03833 0.03239 0.01611 0.01317
 Cumulative Proportion 0.89998 0.93832 0.97071 0.98683 1.00000

First major result figure is the “score plot” of PC1 vs PC2 - how the different candy are related to each other on our new PC axis:

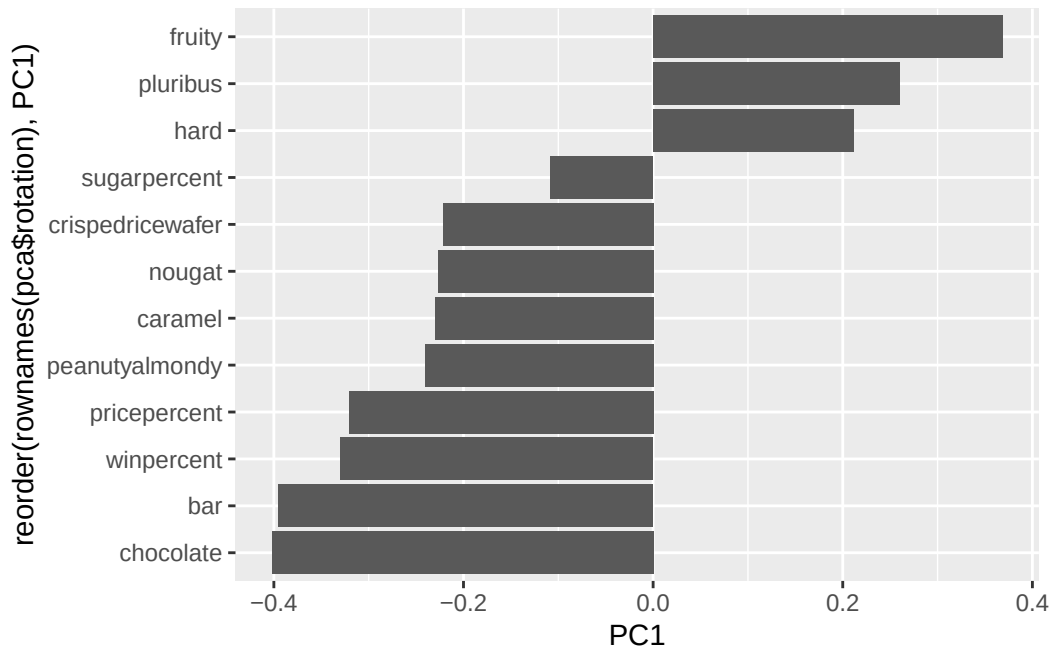
```
ggplot(pca$x) +  
  aes(PC1, PC2, label=row.names(pca$x)) +  
  geom_point(col=mycols) +  
  geom_text_repel(col=mycols)
```



```
library(plotly)  
ggplotly(p)
```

Q24. Complete the code to generate the loadings plot. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

```
ggplot(pca$rotation) +  
  aes(x = PC1, y = reorder(rownames(pca$rotation), PC1)) +  
  geom_col()
```



The variables are picked up strongly by PC1 in the positive direction as this includes fruity, pluribus, and hard since they have large positive loading values. In contrast, variables such as chocolate, bar, winpercent, and pricepercent have strong negative loadings on PC1. This makes sense since chocolate and bar candies tend to be grouped together and are opposite in characteristics to fruity candies. Moreover, the plot displays candies are either chocolate-based or fruity, but not both. This relationship was highlighted previously in the correlation matrix of the corrpplot, where chocolate and fruity were strongly anti-correlated while chocolate and bar were positively correlated.

Summary

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

The “winning” candy seems to be chocolate-based, often a bar, and associated with higher winpercent and pricepercent values. The visualizations also show that many of the highest ranked candies were chocolate candies, while fruity candies tended to rank lower overall. Additionally, the correlation plot shows that chocolate was positively related to popularity and that chocolate and fruity were anti-correlated. Furthermore, the PCA supported this by separating chocolate and bar candies from fruity candies along PC1. Hence, this suggests

that chocolate based candies such as candy bars, tend to be more popular in this dataset as a result.